Usman Ahad

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Education

Lahore University of Management Sciences Lahore, Punjab, Pakistan
Bachelor of Science in Computer Science (CGPA: 3.64 / 4.00) *Aug. 2023 – Present*

- **Honors:** Placed on the Dean's Honour List for consecutive academic years (2023-24, 2024-25, and 2025-26).
- **Graduate Coursework:** Advanced Topics in Machine Learning (CS 6304), AI on Edge Devices (AI 624), Deep Learning (AI 623), Internet of Things (CS 677).
- **Core Coursework:** Foundations of AI & ML, Advanced Topics in Machine Learning, Deep Vision Language Models, AI on Edge Devices

Research Projects

VisConf: Quality-Aware Consensus for Best-of-N VLM Reasoning
PyTorch, Transformers, Qwen2.5-VL, InternVL3, Gemma 4 Jan. 2026 – Sep. 2026

- **Accepted to the NeurIPS 2026 VLM4RWD Workshop.** arXiv
- Developed a training-free Best-of-N framework combining candidate-calibrated Self-Certainty with visual-attention engagement and hyperparameter-free Rank-Weighted Consensus, improving VLM answer selection without external verifier or reward models.
- Evaluated 3 VLMs across MathVista, MMMU-Pro, and MMStar at $N \in \{8, 16, 32\}$; achieved the highest average accuracy at every model-budget setting and outperformed all baselines in **24 of 27** model-dataset-budget configurations.
- Reached **56.36%** mean accuracy, exceeding Self-Consistency by 0.98 points and the strongest baseline by 0.49 points (paired 95% CI: [0.32, 0.66]); delivered gains up to 12.18 points when correct rollouts were present but outnumbered.

PARDA: On-Device Audio Privacy for Smart Glasses using Small Language Models
PyTorch, Qwen3.5-2B, DPO, LS-EEND, ONNX, Raspberry Pi 5 Jan. 2026 – Sep. 2026

- Built PARDA, an IEEE PerCom submission for on-device smart-glasses audio privacy, coupling causal multi-speaker diarization and transcription, persistent disclosure profiles with cross-window retrieval, semantic anonymization, and non-source speech resynthesis.
- Distilled a GPT-5.6 Luna adversary–anonymizer pipeline into Qwen3.5-2B using 30,960 SFT examples and 6,241 DPO preference pairs from 717 CANDOR conversations; reduced leakage by **24.7%** (0.502 → 0.378) while improving utility from 0.875 to 0.900 on 143 held-out conversations.
- Deployed a Q4 model with 0.400 leakage and 0.902 utility; achieved a 0.7703 speaker-mixture-weighted audio-path real-time factor and 23.56 s mean transcription delay across 40 paced Pi 5 replays, while non-source resynthesis drove original-to-anonymized speaker verification to 48.50–50.75% EER.

FedCPR: Contrastive Prototype Rejection for Federated Open-Set Recognition
PyTorch, FedAvg, SupCon, LPR, Energy-Based OOD Detection Nov. 2025 – Feb. 2026

- Developed a federated open-set recognition framework combining supervised contrastive learning, curriculum-based Latent Prototype Rejection (LPR), and energy-based unknown detection under standard FedAvg.
- Generated pseudo-open samples by progressively applying jigsaw, flip, and rotation transformations to intermediate feature maps, reserving low-density regions for unseen classes without raw-image synthesis or additional communication payloads.
- Achieved **85.42 ± 4.02% open-set AUROC** and 90.96 ± 2.82% closed-set accuracy on CIFAR-10 with 6 known and 4 unknown classes across 5 clients, beating PROSER and FedPD by 4.65 and 4.84 AUROC points; LPR added 7.01 points in ablation.

WatchTower: Transformer-Based Network Anomaly Detection for Edge Devices
PyTorch, TFLite, Apache TVM, FedProx, PROSER, Raspberry Pi 5 Aug. 2025 – Dec. 2025

- Built a real-time open-set network intrusion detector for Raspberry Pi 5 using 28 packet-level features, a memory-token Transformer with gated cross-window state, PROSER latent-space outliers, and FedProx for heterogeneous router traffic.
- Achieved **93.66% accuracy** and **92.83% F1** on CIC-IDS2017, improving over the evaluated LSTM-AE baseline by 19.56 and 10.33 percentage points, respectively; validated live DoS detection across 3–8 connected devices.
- Compressed the FP32 model by **67.5%** (4.63 MB → 1.51 MB) through INT8 TFLite quantization, then ran 1,000-trial TVM auto-tuning for ARM Cortex-A76 to reach 3.82 ms single-core latency (260+ packets/sec), a 2.6× speedup.

Teaching Experience

Teaching Assistant for AI on Edge Devices Aug. 2026 – Present
Lahore University of Management Sciences *Lahore, Punjab, Pakistan*

- Support a 15-week graduate course spanning performance measurement, quantization, hardware-aware compilation, TinyML, edge LLMs, and federated privacy across microcontroller, Raspberry Pi, Jetson, and Edge TPU targets.

Teaching Assistant for Deep Learning Jan. 2026 – May 2026
Lahore University of Management Sciences *Lahore, Punjab, Pakistan*

- Supported 100+ MS AI students through programming assignments, technical office hours, and evaluation across CNNs, sequence models, transformers, VAEs, GANs, and diffusion models.

Teaching Assistant for CS 200 (Introduction to Programming) Jan. 2025 – May 2025
Lahore University of Management Sciences *Lahore, Punjab, Pakistan*

- Mentored 200+ undergraduate students in C++ architecture and low-level memory management concepts (pointers, manual allocation).
- Debugged complex student codebases and evaluated algorithmic efficiency, reinforcing core systems programming paradigms.

Technical Skills

Languages: C/C++, Java, SQL, JavaScript, HTML/CSS, MATLAB
Frameworks & Libraries: PyTorch, Transformers, Hugging Face, ONNX Runtime, TFLite, Apache TVM, Scikit-learn, NumPy, pandas, Librosa
Speech & Audio AI: LS-EEND, Streaming ASR, Speaker Embeddings, Speech Separation, Neural TTS, Audio Anonymization
Developer Tools & Platforms: Git, Linux/Unix Shell, Docker, Jupyter Notebooks, Wireshark, Raspberry Pi 5, Edge Hardware
Theoretical Expertise: Open-Set Recognition, Federated Learning (FedProx/FedAvg), Preference Optimization (DPO/LoRA), Multimodal Evaluation (VLMs), Natural Language Processing, Neural Architecture Search, Hardware-Aware Compilation